

Apart from natural language processing – both in terms of understanding and speaking – HAL is also capable of optical recognition with the use of cameras. These two aspects of HAL's abilities unite early in the film to show that he can read lips. In fact, the idea of his ability to read lips is loosely touched upon in another of Clarke's novels, where he explains that the technology isn't too far-fetched. All it would need is a large database of videos of humans talking, with speech. The computer would need to isolate speech and transcribe it, along with associating it with the way the lips move when that word is said. Doing it successfully over several videos would form an index of lip movements that could serve as a visual lexicon to lip-reading.

Bowden and Poole sit in a hatch where HAL can't hear them to talk about him, but HAL manages to read their lips through the glass. He goes on to kill Poole and tries to kill Bowden, who manages to survive and goes on to destroy HAL.

The most astounding part of HAL's artificial intelligence, though, is his social abilities. Clarke's vision of the future of AI had a lot to do with the prevalent work by cognitive scientists of the time. In some way, HAL is the extreme version of what it takes to pass the Turing Test – an intelligent agent that is almost sentient.

HAL displays remarkable social understanding of humanity and over the course of the film, displays several traits such as self-defence instincts, humour, sarcasm, social negotiation and fear. In the final moments, when Bowden is dismantling HAL, he's shown to be pleading against his demise,



KITT from *Knight Rider*